

6-Bit R-String DAC

EE 435 Final Project Report

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1. Introduction

1.1 Project Objective

The goal of this project was to design a 6-bit digital-to-analog converter (DAC) in TSMC 180nm that achieves an effective number of bits (ENOB) of at least 8 from an INL viewpoint, with a yield of at least 90% over 200 Monte Carlo samples considering both mismatch and process variations. No ideal components were permitted except voltage sources.

The ENOB requirement means that although the DAC has only 64 physical output codes (6 bits of resolution), its integral nonlinearity must be tight enough to correspond to 8-bit precision. Specifically, INL must remain below 0.5 LSB evaluated at 8-bit resolution, which translates to 0.125 LSB at the 6-bit physical scale. This is a factor of 4 tighter than what a standard 6-bit DAC would require.

1.2 Architecture Selection

An R-string (resistor-ladder) architecture was chosen. In this topology, a series string of unit resistors divides a reference voltage into equally-spaced tap voltages. A set of switches, controlled by a decoder, selects one tap and passes its voltage through an output buffer to the load.

The R-string was selected over alternatives such as R-2R ladders, current-steering DACs, and charge-redistribution DACs for several reasons. First, the R-string is inherently monotonic: because each tap voltage is the sum of series resistor drops from ground, adding any positive resistance always increases the tap voltage. This guarantees $DNL > -1$ LSB by construction, eliminating an entire class of potential failures. Second, the static INL of an R-string DAC depends almost exclusively on resistor matching. Switch on-resistance and op-amp nonidealities contribute negligibly to static INL because the output buffer draws little current through the switches. This means the Pelgrom mismatch model can be used to quite effectively to predict yield from the resistor area alone, simplifying the design process significantly. The main downside as far as I could tell was significant layout work would be required, but because of no area constraints, and a relatively small total number of bits, it felt doable.

1.3 8×8 Row/Column Organization

The 64 unit cells were organized as an 8-by-8 array with a row/column decoder scheme rather than a flat 6-to-64 decoder. Two 3-to-8 decoders generate 8 row select lines and 8 column select lines respectively. At each cell in the array, a 2-input NAND gate combines the corresponding row and column signals to produce a cell-specific select signal.

This approach reduces the number of digital wires routed across the array from 64 (for a flat decoder) to just 16 (8 row + 8 column). It also allows smaller logic gates (NAND3 instead of NAND6), and the regular 2D structure maps is fairly easy to layout.

1.4 Summary of Results

Parameter	Specification	Achieved	Margin
Physical Resolution	6 bits (64 codes)	6 bits	—
ENOB (INL-based)	≥ 8 bits	Mean: 10.1 bits	+2.1 bits
Worst-Case ENOB	≥ 8 bits	8.69 bits	+0.69 bits
INL _{max}	< 0.125 LSB	Mean: 30.7 mLSB	1.9x
DNL _{max}	< 0.25 LSB	Mean: 15.0 mLSB	4.2x
MC Yield (200-pt)	$\geq 90\%$	100% (200/200)	—

2. Unit DAC Cell

The unit DAC cell is the fundamental building block of the design. All 64 instances in the array are identical; differentiation between cells comes entirely from external wiring (which row and column lines connect to each cell). Each cell contains all the components needed to contribute one tap to the resistor string and to conditionally pass that tap voltage to the output bus.

2.1 Cell Contents and Function

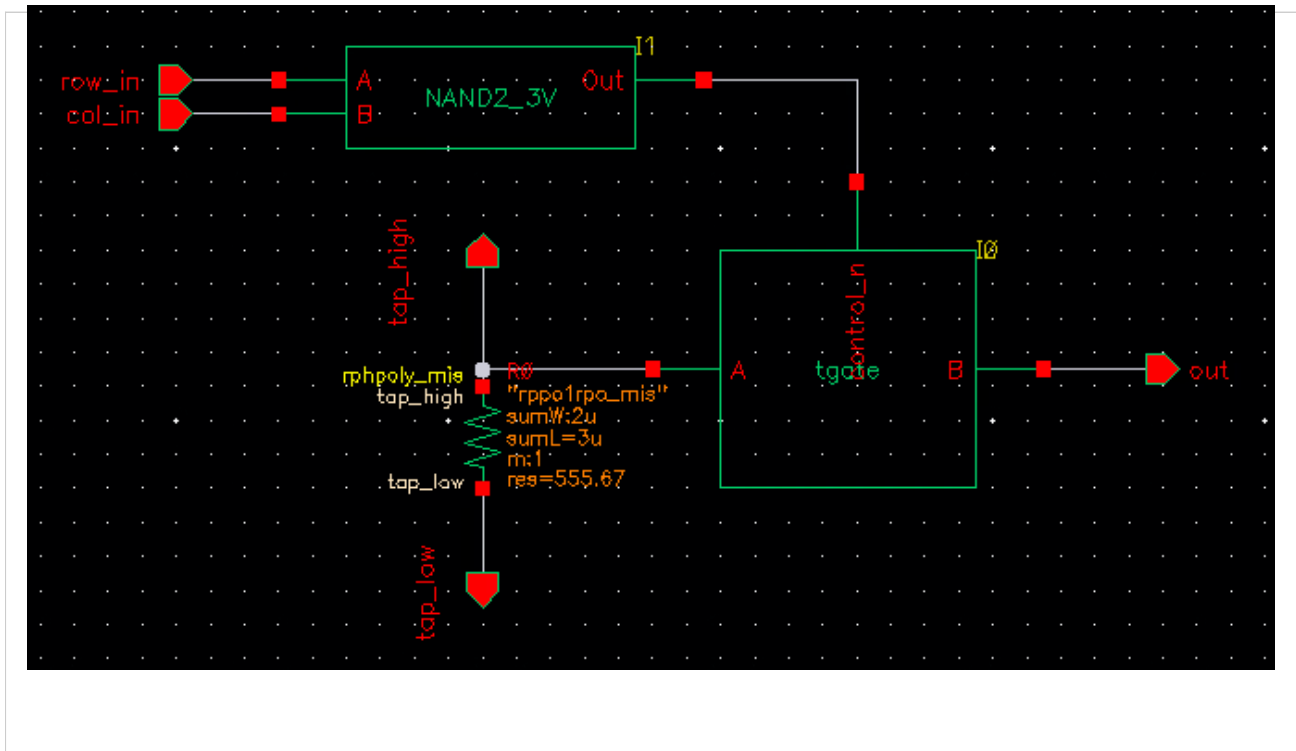


Figure: Unit cell schematic

Each unit cell contains one unit resistor, one 2-input NAND gate, one inverter, and one transmission gate (T-gate), for a total of 8 transistors and 1 resistor per cell.

The unit resistor connects between the cell's `tap_lo` and `tap_hi` pins. When 64 cells are chained together (cell *k*'s `tap_hi` connected to cell *k*+1's `tap_lo`), the resistors form a continuous series string from ground to V_{REF} . The voltage at each tap is proportional to the number of unit resistors below it.

The NAND2 gate serves as a combiner for the row/column decoder outputs. It takes two inputs: row_in from the row decoder and col_in from the column decoder. Its output is LOW only when both inputs are HIGH, which occurs for exactly one cell in the array per code. This implements the logical AND of the row and column selection.

The inverter generates the complementary control signal needed by the T-gate. Because the NAND2 naturally produces an active-low output (LOW when selected), the T-gate was designed with active-low control to eliminate the need for an extra inverter. The NAND2 output drives the PMOS gate of the T-gate directly (PMOS turns on when its gate is LOW), while the inverter output drives the NMOS gate (NMOS turns on when its gate is HIGH). This optimization saves 2 transistors per cell compared to using a separate inverter before an active-high T-gate, amounting to 128 transistors saved across the full 64-cell array.

The T-gate is a complementary transmission gate consisting of one NMOS and one PMOS transistor in parallel. When both are turned on (cell selected), the tap_hi voltage passes through to the vout_out pin with low on-resistance. When both are off (cell not selected), the switch presents a very high impedance, effectively isolating this cell from the output bus. All 64 cells' vout_out pins are connected to a single common node, forming a wired-OR multiplexer where only the selected cell drives the output.

2.2 Signal Flow Example

Consider code = 13 (binary 001101). The dec2bin6 stimulus block produces b5:b0 = 001101. The column decoder receives bits 3-5 (001) and asserts col<1> HIGH. The row decoder receives bits 0-2 (101) and asserts row<5> HIGH. At cell (row 5, col 1), the NAND2 sees both inputs HIGH, so its output goes LOW. This LOW signal drives the PMOS gate of the T-gate (turning it on) and feeds through the inverter to produce a HIGH signal on the NMOS gate (also turning it on). The T-gate conducts, connecting tap<14> to the common output bus. All other cells have at least one NAND2 input LOW, so their T-gates remain off and do not interfere with the output.

2.3 Component Sizing

Component	W	L	Notes
Unit R (rphpoly_mis)	2 μm	3 μm	$\sim 555 \text{ kohm}$, area = 6 μm^2
T-gate NMOS (nch3_mis)	2 μm	350 nm	R_ON $\sim 400 \text{ ohm}$ at rails
T-gate PMOS (pch3_mis)	6 μm	350 nm	3x NMOS for mobility balance
NAND2 NMOS (x2)	1 μm	350 nm	2x min for series stacking
NAND2 PMOS (x2)	1 μm	350 nm	Parallel, standard sizing
INV NMOS	0.5 μm	350 nm	Minimum practical width
INV PMOS	1 μm	350 nm	2x NMOS for balanced drive

All transistors use 3.3V I/O devices from the TSMC 180nm PDK with _mis model variants to enable Monte Carlo mismatch simulation.

2.4 Cell Pins

Pin	Direction	Description
tap_hi	inputOutput	Top of unit resistor; also connects to T-gate input (A)
tap_lo	inputOutput	Bottom of unit resistor; chains to adjacent cell's tap_hi
vout_out	inputOutput	T-gate output (B); all 64 cells share this common bus
row_in	input	Row select line from row decoder
col_in	input	Column select line from column decoder

Power (vdd!) and ground (gnd!) are global nets, not explicit pins on the cell symbol.

2.5 Testbench

The unit cell was verified in a standalone testbench before integration into the array. A 1.5V DC source drove the tap_in terminal, and a 1 uA current source provided a measurement load on the vout_out terminal. Row_in and col_in were driven through all four combinations: (0,0), (0,1), (1,0), and (1,1). Only the (1,1) combination produced V_OUT equal to the tap voltage (within the R_ON voltage drop of a few hundred microvolts). All other combinations produced a high-impedance output, confirming correct switch behavior.

2.6 Layout

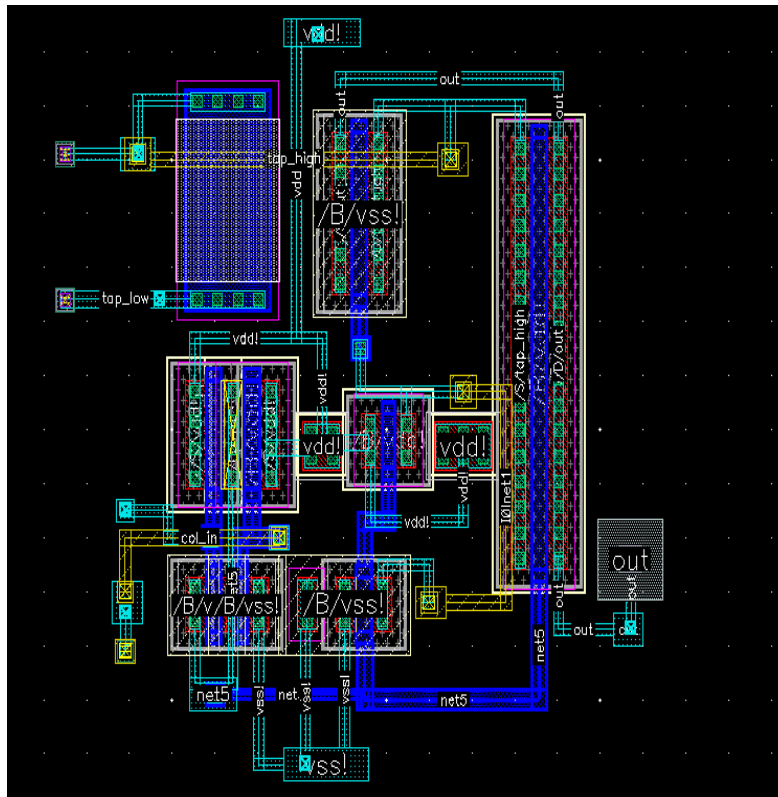
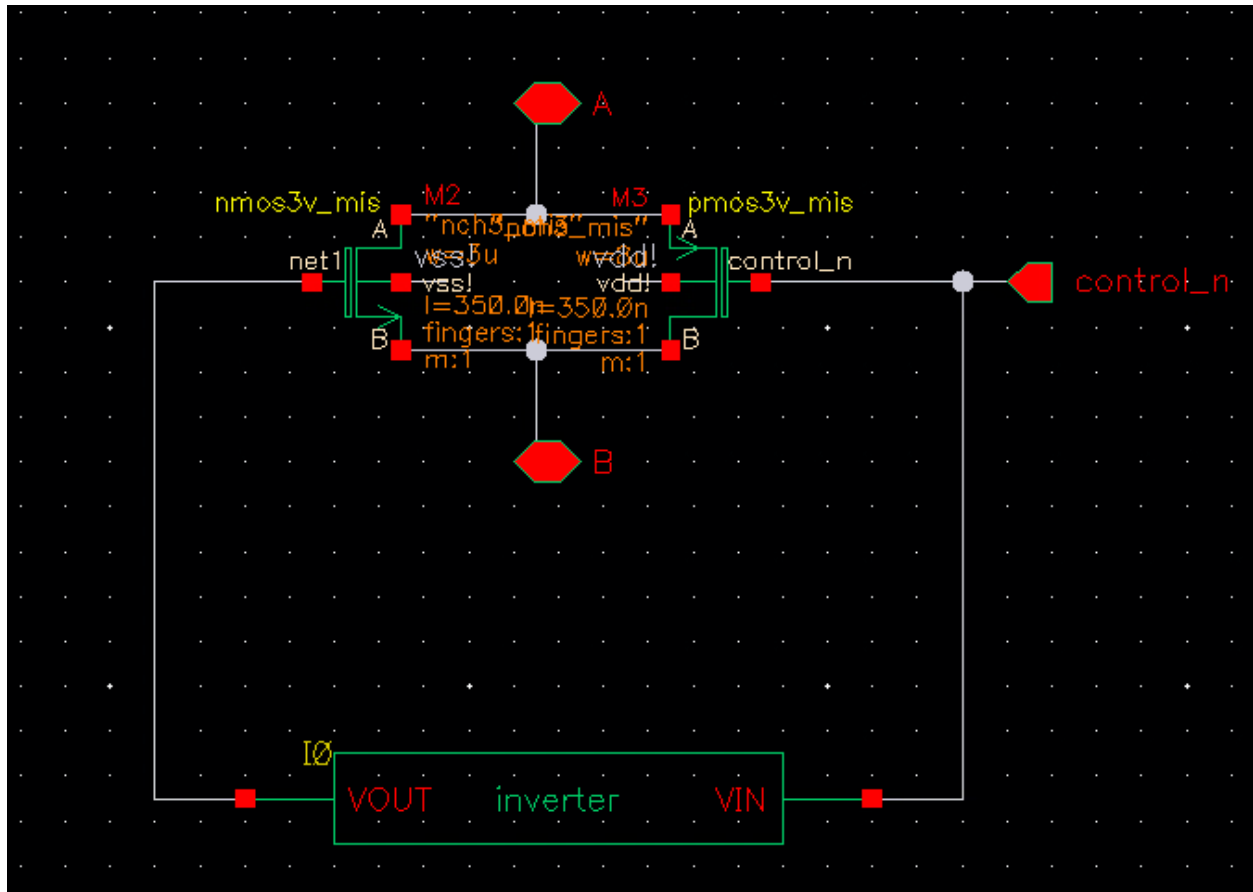


Figure: Unit cell layout

The layout is a little bit messy, but the main things to point towards are the relatively small resistor in the top left corner, the layout was only really optimized for making an array of them as simple to wire as possible.

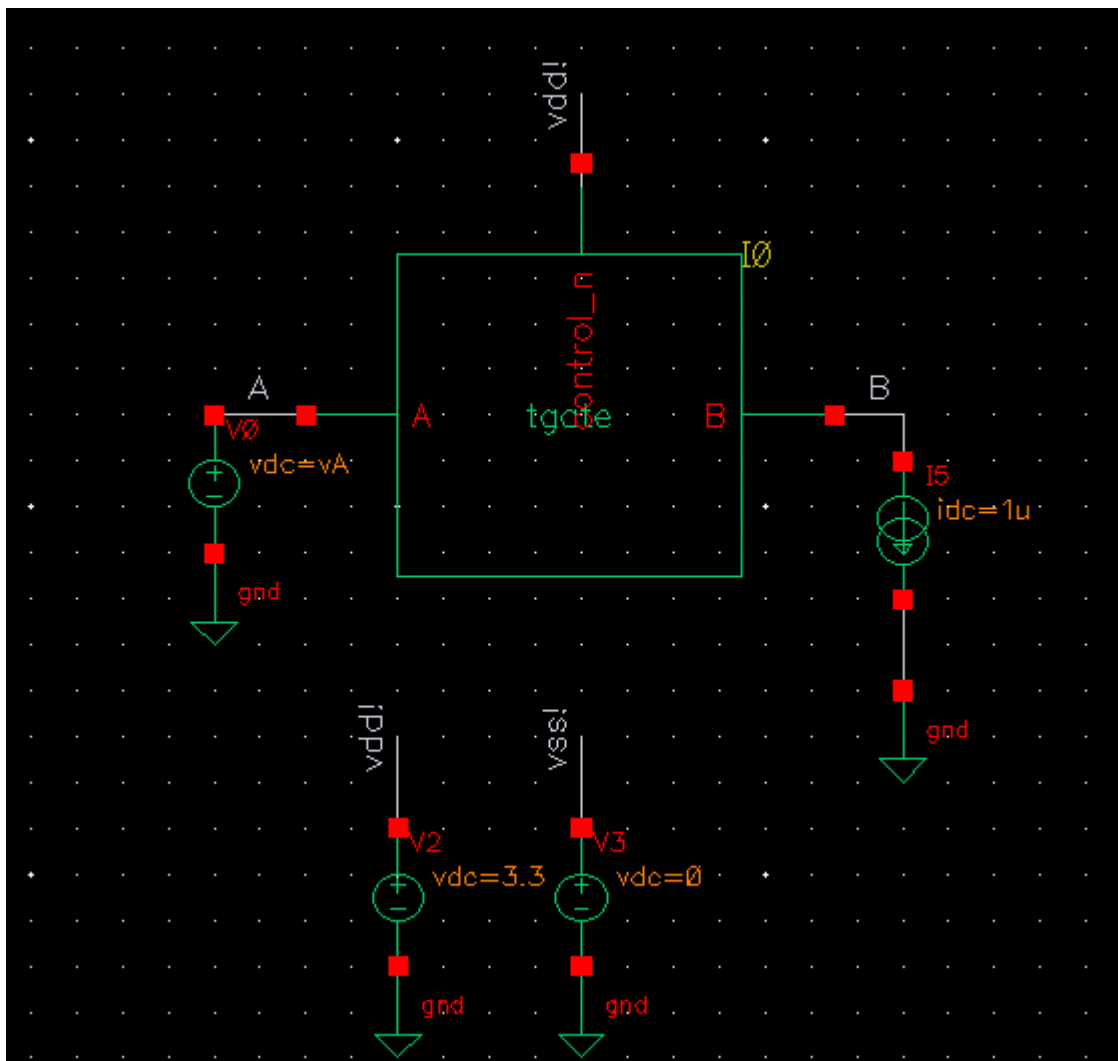
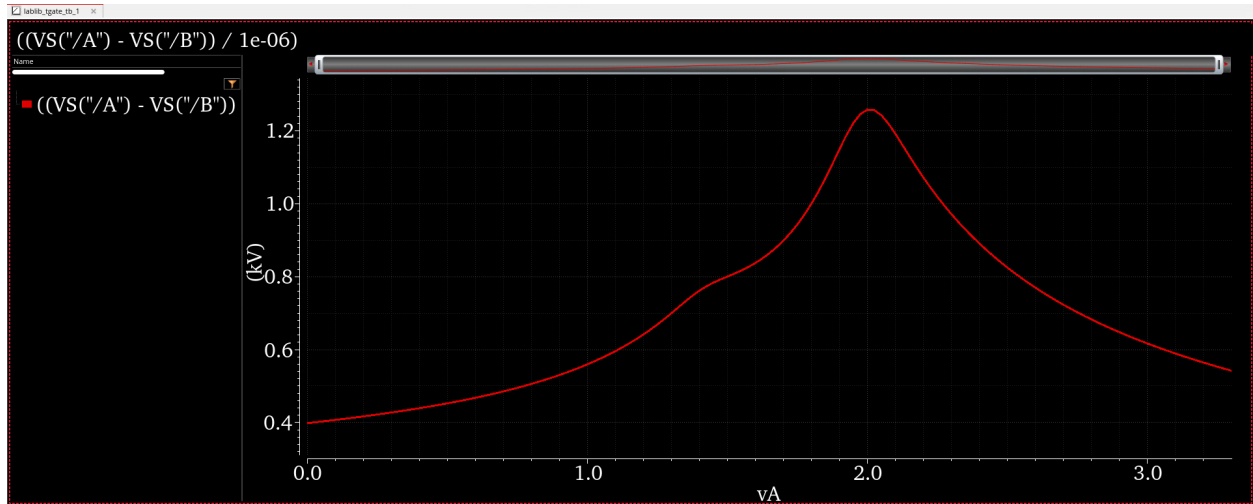
3. T-Gate Switch Characterization



3.1 Design

The transmission gate uses a complementary NMOS/PMOS pair. The NMOS ($W=2\mu\text{m}$, $L=350\text{nm}$) provides low on-resistance for signal voltages near ground, while the PMOS ($W=6\mu\text{m}$, $L=350\text{nm}$) provides low on-resistance for signal voltages near VDD. The PMOS is sized 3x wider than the NMOS to partially compensate for the lower hole mobility, yielding a more uniform R_{ON} across the signal range.

3.2 R_ON Characterization



Figures: tgate testbench schematic, and results

The T-gate's on-resistance was characterized by sweeping the input voltage (V_A) from 0 to 3.3V while passing a 1 μ A test current through the switch. The control signal was held at VDD (always on). R_{ON} was computed as $(V_A - V_B) / 1 \mu A$.

The measured R_{ON} ranged from approximately 400 ohms at the supply rails to a peak of about 1.25 kohm near $V_A = 2V$. The peak occurs where both transistors have reduced gate overdrive due to body effect. Despite this variation, R_{ON} has zero effect on static DAC linearity because the op-amp input presents essentially infinite impedance. The current through the switch in normal operation is in the picoamp range, producing a voltage drop of picovolts — completely negligible compared to the ~ 16 mV LSB.

3.3 Gate Polarity Issue

During initial testing, the T-gate was found to conduct when the control signal was LOW and disconnect when it was HIGH — the opposite of the intended behavior. Investigation revealed that the internal inverter was driving the wrong transistor: the NMOS gate was receiving the inverted control while the PMOS gate received the direct control. Since PMOS turns on with a LOW gate and NMOS turns on with a HIGH gate, this wiring made the switch active-low by accident.

The fix was to swap the gate connections: route the control signal directly to the PMOS gate and the inverter output to the NMOS gate. This deliberately made the T-gate active-low, which proved advantageous when combined with the NAND2 combiner (whose output is naturally active-low when both inputs are high). The swap eliminated the need for an additional inverter between the NAND2 and the T-gate.

4. Resistor String Design

4.1 Resistor Selection

The unit resistor uses the rphpoly_mis model from the TSMC 180nm PDK, which is a P+ polysilicon resistor with mismatch characterization. The sheet resistance is approximately 311 ohms per square. The Pelgrom area matching coefficient, extracted from the PDK, is $A_{PEL,R} = 1.146$ percent-micrometers.

4.2 Pelgrom Extraction Method

Before sizing the unit resistor, the Pelgrom area matching coefficient ($A_{PEL,R}$) was extracted empirically from the PDK models using a standalone testbench. A single rphpoly_mis resistor was placed in a testbench with a known DC current flowing through it. A 200-point Monte Carlo simulation was run with mismatch enabled and process variation disabled, so the only source of variation was the resistor's statistical mismatch model. The resistance value was extracted at each MC iteration by dividing the measured voltage across the resistor by the applied current. This was repeated at several different sizes and then calculated with

$$A_{PEL} = \sigma_{R/R} * \sqrt{W * L}$$

4.3 Sizing Derivation

The matching requirement was derived from the ENOB target. For 90% yield at 8-bit ENOB on a 64-element string, the maximum allowable sigma of INL_max is approximately 0.04 LSB (assuming the spec of 0.125 LSB corresponds to roughly a 3-sigma event). The relationship between per-resistor mismatch and INL is $\sigma_{INL_max} = \sigma_{R/R}$ times the square root of N divided by 2, where N = 64 is the number of unit resistors. This gives $\sigma_{R/R}$ less than approximately 1 percent.

Applying the Pelgrom model, the minimum resistor area is $(A_{PEL} / \sigma_{target})^2 = (1.146 / 1.0)^2 = 1.31 \text{ um}^2$. The chosen sizing of W = 2 um, L = 8.92 um provides an area of 17.84 um², which is over 13 times the minimum required. This large margin was deliberate, providing headroom for layout-induced gradient effects and post-extraction parasitic degradation. The

resulting $\sigma_{R/R}$ is 0.27%, giving a predicted σ_{INL_max} of 0.011 LSB — more than 11 times below the 0.125 LSB specification.

4.4 Pelgrom Verification: The Square-Root-of-Two Test

To verify that Monte Carlo mismatch was being correctly applied to all 64 resistor instances independently, a controlled experiment was performed. The full 200-point MC simulation was run at the original resistor sizing, then repeated with the resistor area halved (L reduced from 8.92 μm to approximately 4.46 μm). The Pelgrom model predicts that halving the area should increase $\sigma_{R/R}$ by around the square root of 2, and consequently all INL and DNL statistics should scale by the same factor.

Metric	Original (L=8.92 μm)	Halved (L~4.46 μm)	Ratio
Mean INL_max	21.0 mLSB	29.7 mLSB	1.41x
Sigma INL_max	7.12 mLSB	10.07 mLSB	1.41x
Max INL_max	48.9 mLSB	69.2 mLSB	1.41x
Mean DNL_max	8.36 mLSB	11.82 mLSB	1.41x

Every metric scaled by exactly 1.41 (the square root of 2), confirming the model we were using in design was working as expected. Although in practice to pick my final resistor size of 2W, 3L I just cut length until the ENOB had a solid margin of error over the requirement.

5. Row/Column Decoder

5.1 Architecture

Each 3-to-8 decoder consists of 3 inverters (to generate bit complements), 8 NAND3 gates (one per output), and 8 output inverters (to convert the NAND3 active-low outputs to active-high select signals). The row decoder receives the 3 least-significant bits of the code (b0, b1, b2) and the column decoder receives the 3 most-significant bits (b3, b4, b5). This bit assignment was determined empirically, the unit cell bus mapping required rows to correspond to LSBs and columns to MSBs.

5.2 NAND3 Wiring Pattern

Each NAND3 gate is wired to a unique combination of true and complement input bits, such that its output is asserted for exactly one of the 8 possible 3-bit input codes. For output sel<k>, bit position i receives the true bit if bit i of k is 1, or the complement if bit i of k is 0.

Output	Binary	Input A	Input B	Input C
sel<0>	000	A_bar	B_bar	C_bar
sel<1>	001	A_bar	B_bar	C
sel<2>	010	A_bar	B	C_bar
sel<3>	011	A_bar	B	C
sel<4>	100	A	B_bar	C_bar
sel<5>	101	A	B_bar	C
sel<6>	110	A	B	C_bar
sel<7>	111	A	B	C

5.3 Gate Sizing

Gate	NMOS W	PMOS W	L
INV	0.5 um	1.0 um	350 nm
NAND2 (in unit cell)	1.0 um	1.0 um	350 nm
NAND3 (in decoder)	1.5 um	1.0 um	350 nm

NMOS widths are scaled proportionally to the number of series-stacked transistors (1x for INV, 2x for NAND2, 3x for NAND3) to maintain comparable pull-down strength. PMOS transistors are in parallel and do not require width scaling. All gates use 3.3V mismatch devices.

5.4 Layout

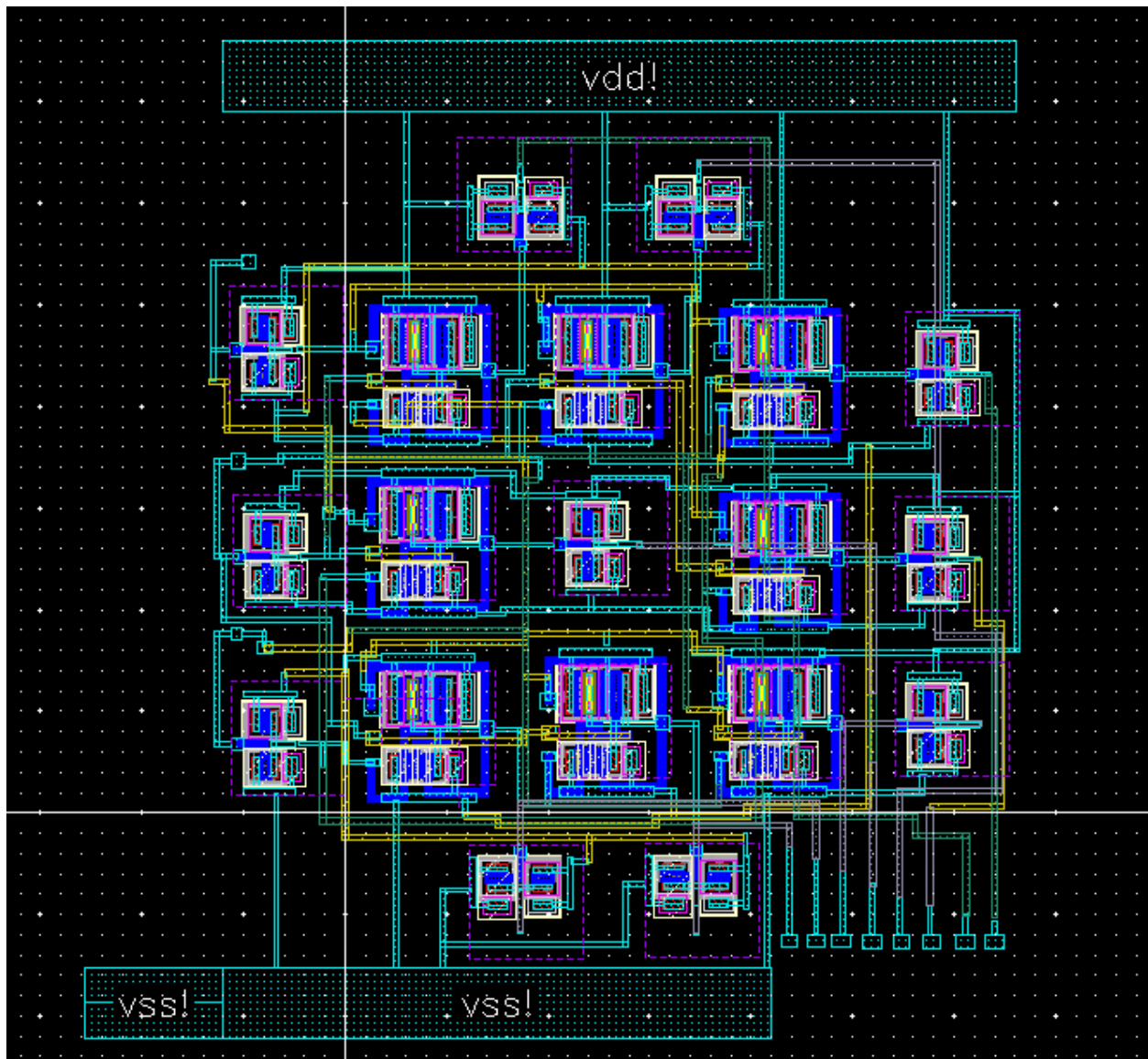


Figure: Decoder layout

The decoder layout was again somewhat messy, the main things to note are that I tried to achieve relatively even distribution of devices, and the outputs in the bottom right hand corner. But everything passed lvs and drc.

6. Op-Amp

6.1 Topology

The op-amp is a 7-transistor two-stage Miller-compensated operational amplifier with an NMOS input differential pair. The first stage consists of the NMOS input pair with a PMOS current-mirror active load, biased by a PMOS tail current source. The second stage is a common-source NMOS amplifier with a PMOS current-source load. Miller compensation is provided by a MiM capacitor (C_c) in series with a nulling resistor (R_c) connected between the first-stage output and the second-stage output. The opamp also uses a widlar current reference which will be described in more detail later. The op-amp operates on a plus/minus 1.65V dual supply for a total of 3.3V across the rails. It is essentially the lab5 opamp with current reference.

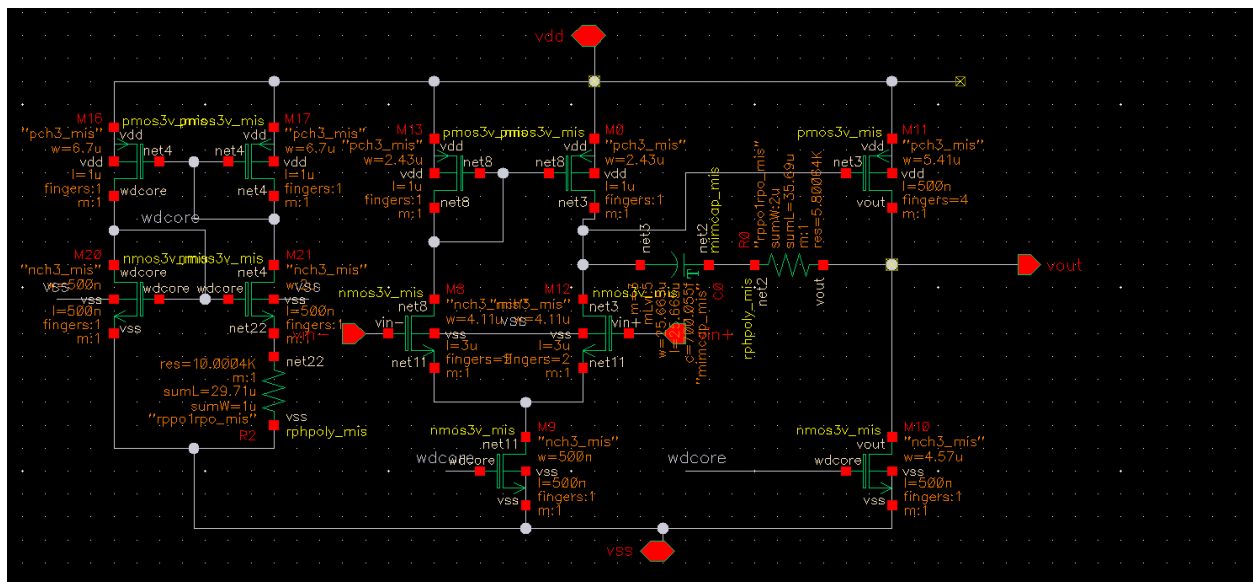


Figure: Op-amp /w Widlar schematic

6.2 Performance

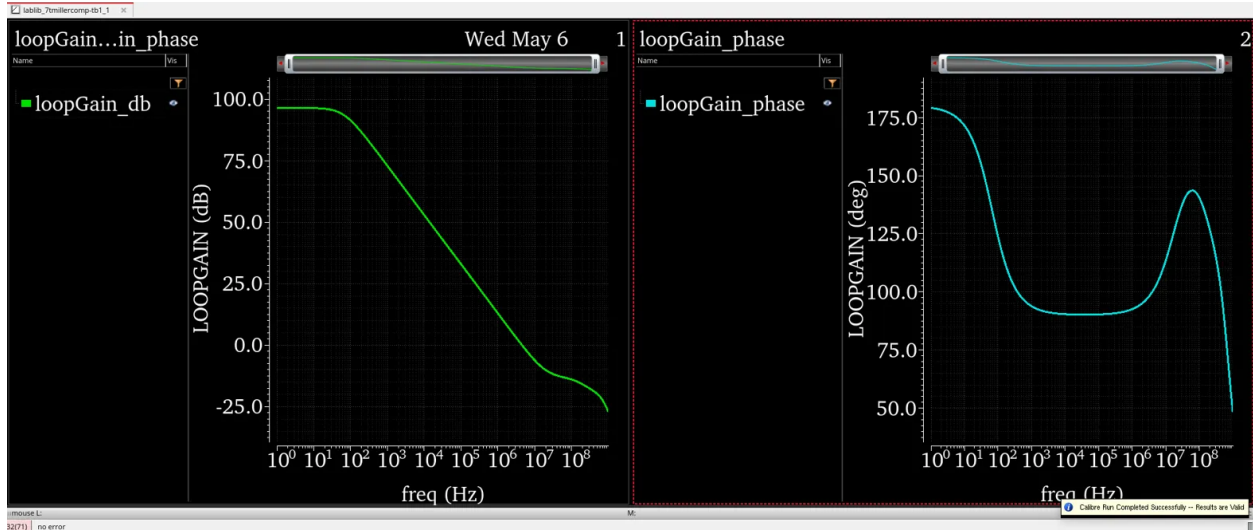


Figure: Loop gain magnitude and phase Bode plots

Parameter	Value
DC Open-Loop Gain	~100 dB
Unity-Gain Bandwidth	~2 MHz
Phase Margin	101 degrees
Supply	+/- 1.65V (3.3V total)

The DC gain of 100 dB provides 30 dB of margin over the 70 dB requirement for sub-LSB gain error. We can see the results aren't particularly good, but I essentially stopped trying to optimize the op-amp as soon as it was able to pass specification.

6.3 Compensation

The Miller capacitor C_c creates the dominant pole, pushing it to low frequencies to ensure a -20 dB/decade rolloff through the unity-gain crossover. The nulling resistor R_c cancels the right-half-plane zero that Miller compensation introduces. R_c was sized to approximately $1/g_m$ of the second-stage transistor. The compensation values are $C_c = 2$ pF (MiM capacitor, `mimcap_mis` model) and $R_c =$ approximately 1 kohm (`rphpoly_mis`).

6.4 Layout

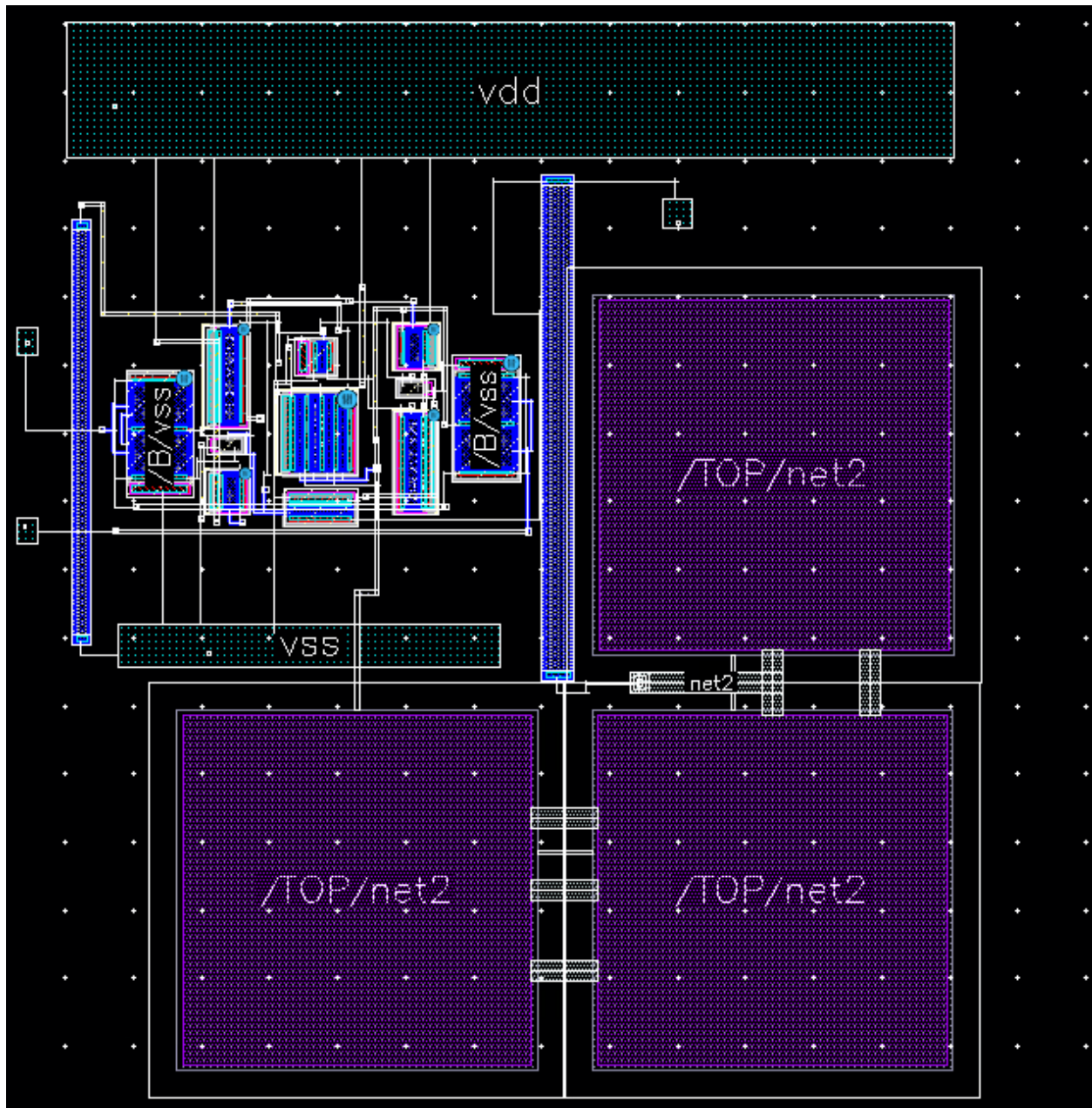


Figure: Op-amp w/Widlar layout

The op-amp layout passed DRC and LVS. The MiM compensation capacitor dominates the layout area.

7. Widlar Current Reference

7.1 Topology

The bias generator is a self-biased current reference based on the Widlar topology. It consists of a cross-coupled PMOS mirror pair (MP1 diode-connected, MP2 mirror slave) and a cross-coupled NMOS pair (MN1 slave, MN2 diode-connected with a degeneration resistor R_{REF} in its source). The PMOS gates share one common node (pgate) and the NMOS gates share another (ngate). The two mirror loops constrain each other, forcing the circuit to settle at a unique operating current determined by R_{REF} and the W/L ratio between MN1 and MN2.

7.2 Bias Stability Issue and Fix

The initial Widlar design caused the op-amp to fail on approximately 15% of Monte Carlo iterations. The INL histogram was distinctly bimodal: about 170 iterations showed clean behavior (INL_max around 30 mLSB), while roughly 30 iterations showed catastrophic failure (INL_max of 5 to 20 LSB) where the op-amp completely lost its operating point.

The root cause was identified as insufficient R_{REF} . With a small degeneration resistor, the reference current was too sensitive to threshold voltage variation across process corners. At slow corners (where V_{TH} increases), the Widlar's current dropped so far that the op-amp could no longer maintain adequate gain, causing its output to rail and the DAC transfer curve to collapse. The fix involved two changes: increasing R_{REF} to stabilize the current reference against V_{TH} variation, and widening the second-stage input transistor to maintain output drive at the lower bias currents. After these changes, the bimodal distribution disappeared entirely and the INL histogram became a clean Gaussian, yielding 100% pass rate across all 200 MC iterations.

I think there is still work to be done on biasing to get better results, but after passing specification within simulation I chose to move onto layout in order to get everything done.

8. Full DAC Integration and Testbench

8.1 Testbench Structure

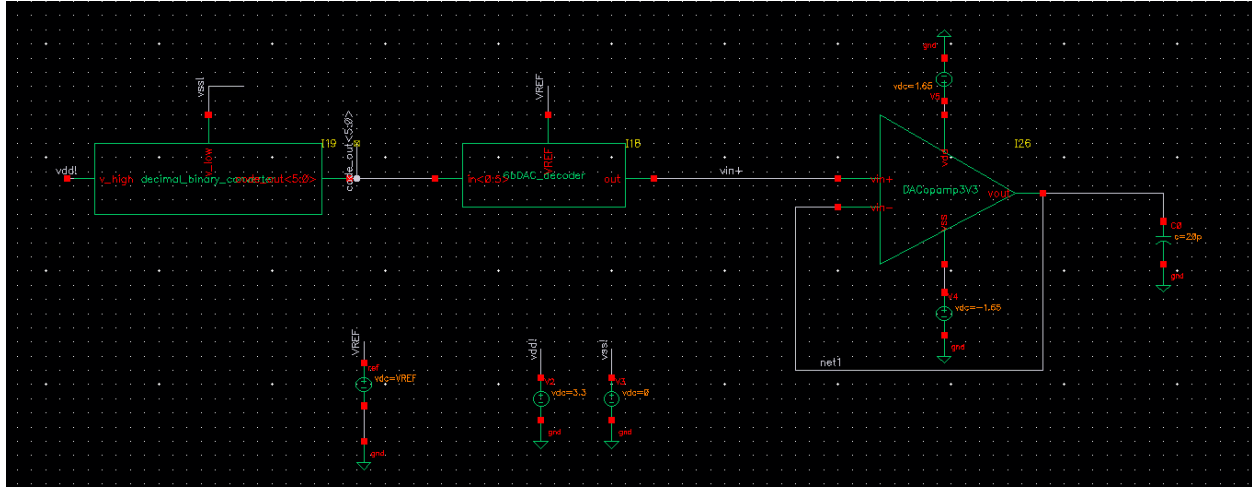


Figure: Top-level testbench schematic showing dec2bin6, DAC block, op-amp, supplies

The top-level testbench consists of the dec2bin6 Verilog-A stimulus block (which converts an integer code parameter into 6 digital outputs with realistic transition edges), the DAC block (containing both 3-to-8 decoders and the 8x8 unit cell array), and the op-amp configured as a unity-gain buffer with output feedback to vin-. The dec2bin6 block is the only Verilog-A component in the design and serves purely as a stimulus generator, it is not part of the physical chip.

8.2 Bus Notation and Array Wiring

The 64 unit cells are instantiated using Cadence bus notation as unit_cell<0:63>. The tap_lo pin is wired to the bus expression gnd,tap<1:63> and tap_hi to tap<1:63>,VREF. This creates a chain of 64 resistors in series: cell 0's resistor spans from ground to tap<1>, cell 1's from tap<1> to tap<2>, and so on, with cell 63's resistor ending at VREF.

The row_in pin uses a comma-separated bus expression that maps each cell index k to row<k mod 8>, repeating each row value 8 times. Similarly, col_in maps to col<k div 8>. This ensures that incrementing the code by 1 moves to the adjacent cell in the array, producing a monotonically increasing tap voltage.

8.3 Nominal DC Sweep

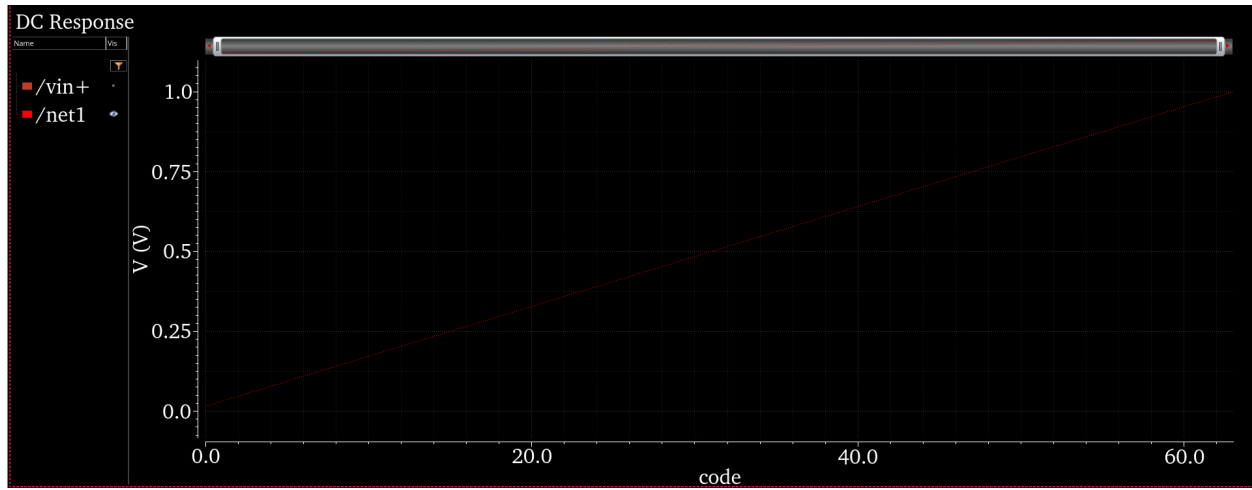


Figure: Nominal DC sweep: V_{OUT} vs code (0-63), clean linear staircase

A parametric DC sweep of the code variable from 0 to 63 produces a clean linear staircase from approximately 0V to VREF (1.0V). The output is monotonic across all 64 codes with INL_{max} below 1 mLSB at the nominal corner, confirming that the digital decoder chain, switch array, and op-amp buffer are all functioning correctly.

8.4 VREF Selection

The reference voltage was progressively lowered from the initial 1.6V to its final value of 1.0V. At $VREF = 1.6V$, the op-amp output saturated at the top codes because V_{OUT} could not reach the positive supply rail. Systematic error analysis (plotting the mean INL across all MC iterations) revealed a deterministic bow at codes 245-255 that was not caused by mismatch but by the op-amp output stage running out of headroom. Lowering VREF to 1.3V resolved the saturation for the 8-bit (256-code) version, and further reduction to 1.0V was adopted for the final 6-bit (64-code) version to ensure adequate op-amp headroom across all process corners.

8.5 Model Library Configuration

File	Section	Purpose
cor_std_mos.scs	tt	Standard MOS typical corner
rf018.scs	mc_3v	3V transistor mismatch
rf018.scs	mc_res	Resistor MC (1 of 2)
rf018.scs	stat_res	Resistor statistical (2 of 2)
rf018.scs	mc_mim	MiM cap MC (1 of 2)
rf018.scs	stat_mim	MiM cap statistical (2 of 2)

The cor_3v.scs file (section tt_3v) was disabled because it conflicts with the mc_3v section of rf018.scs, both define the nch3_mis and pch3_mis subcircuits, causing a duplicate definition error. Per TA guidance, both mc_res and stat_res are required simultaneously for resistors to exhibit stochastic behavior in Monte Carlo simulation. All device instances throughout the design use _mis model variants.

9. Monte Carlo Results

9.1 Configuration

Monte Carlo analysis was run with 200 samples, with both mismatch and process variation enabled. The analysis consisted of a DC parametric sweep of the code variable from 0 to 63 (step 1) at each MC iteration. Three output expressions were evaluated per iteration: INL_max, DNL_max, and ENOB.

9.2 Calculator Expressions

Note /net1 is output

INL_max was computed using the Cadence calculator expression: $\text{ymax}(\text{abs}((\text{VS}("/\text{net1}") - \text{value}(\text{VS}("/\text{net1}") 0) - ((\text{value}(\text{VS}("/\text{net1}") 63) - \text{value}(\text{VS}("/\text{net1}") 0)) / 63) * \text{xval}(\text{VS}("/\text{net1}")))) / ((\text{value}(\text{VS}("/\text{net1}") 63) - \text{value}(\text{VS}("/\text{net1}") 0)) / 63)))$. This implements endpoint-fit INL in physical LSB units: the ideal line is defined by V_OUT at code 0 and code 63, and INL at each code is the deviation from this line divided by LSB.

DNL_max was computed as: $\text{ymax}(\text{abs}(\text{deriv}(\text{VS}("/\text{net1}")) / ((\text{value}(\text{VS}("/\text{net1}") 63) - \text{value}(\text{VS}("/\text{net1}") 0)) / 63) - 1))$. This computes the successive code-to-code step size normalized by the ideal LSB, minus 1.

9.3 INL Results

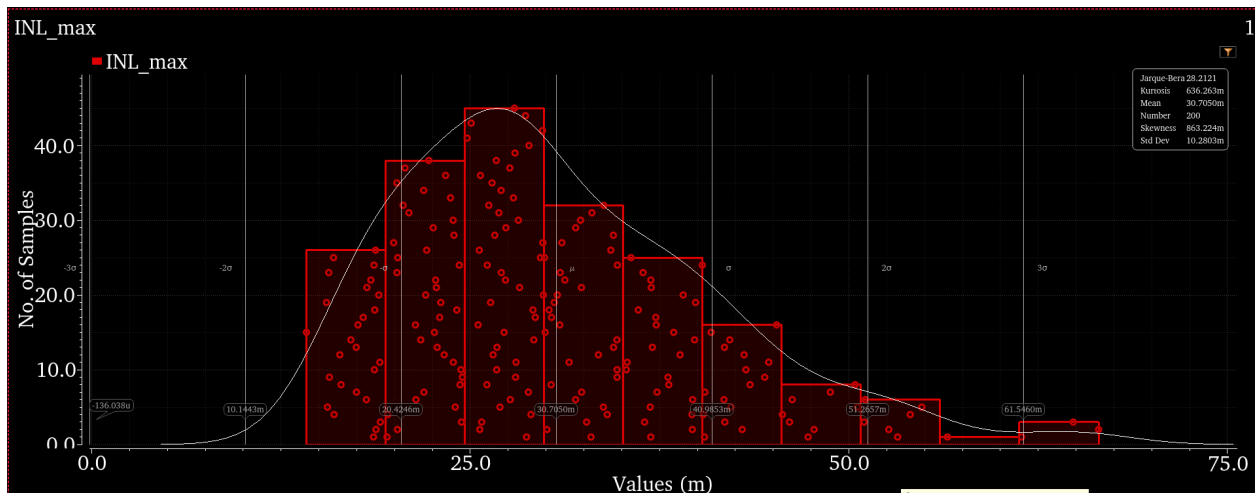


Figure: INL_max histogram from Cadence Monte Carlo results (200 iterations)

The INL_{max} distribution is unimodal with a mean of 30.7 mLSB and a worst case of approximately 65 mLSB. The specification of 0.125 LSB (125 mLSB) is indicated by a vertical line on the histogram; all 200 iterations fall well below this limit. The distribution shape is consistent with the theoretical Rayleigh distribution expected for the maximum of correlated Gaussian random variables (the INL at each code is a random walk of resistor mismatches).

9.4 DNL Results

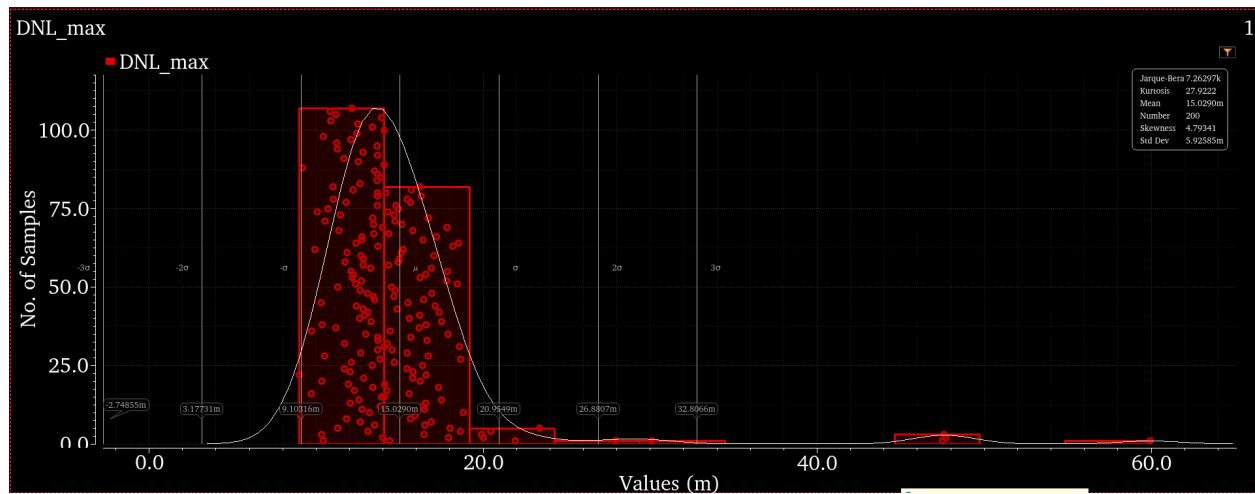


Figure: DNL_{max} histogram from Cadence Monte Carlo results (200 iterations)

The DNL_{max} distribution has a mean of 15.0 mLSB and a worst case of approximately 60 mLSB, well below the 0.25 LSB (250 mLSB) specification. The R-string topology guarantees DNL greater than -1 LSB (inherent monotonicity), so the DNL specification is met by construction in addition to being verified by simulation.

9.5 ENOB Results

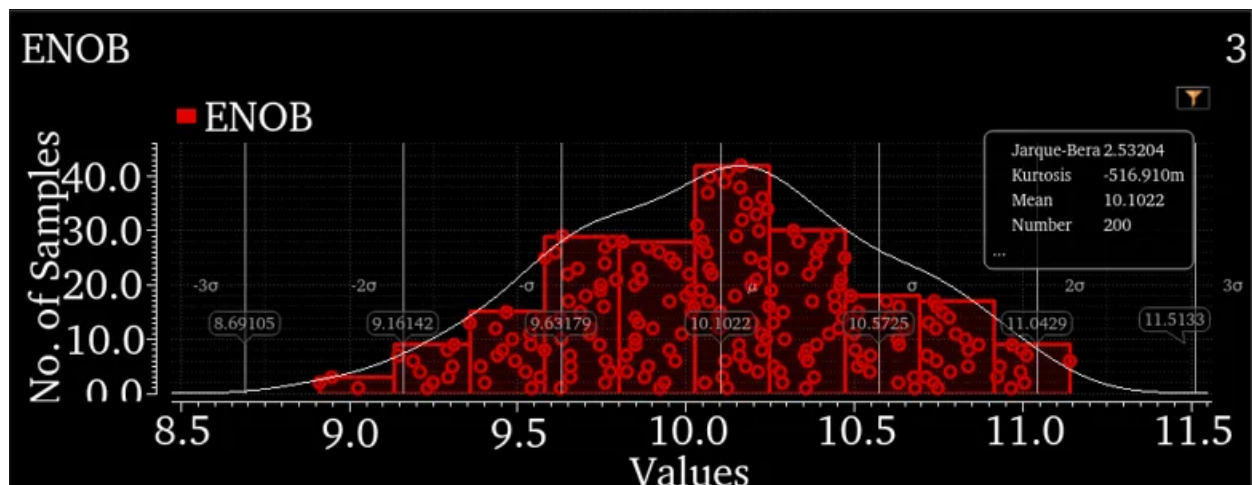


Figure: ENOB histogram from Cadence Monte Carlo results (200 iterations)

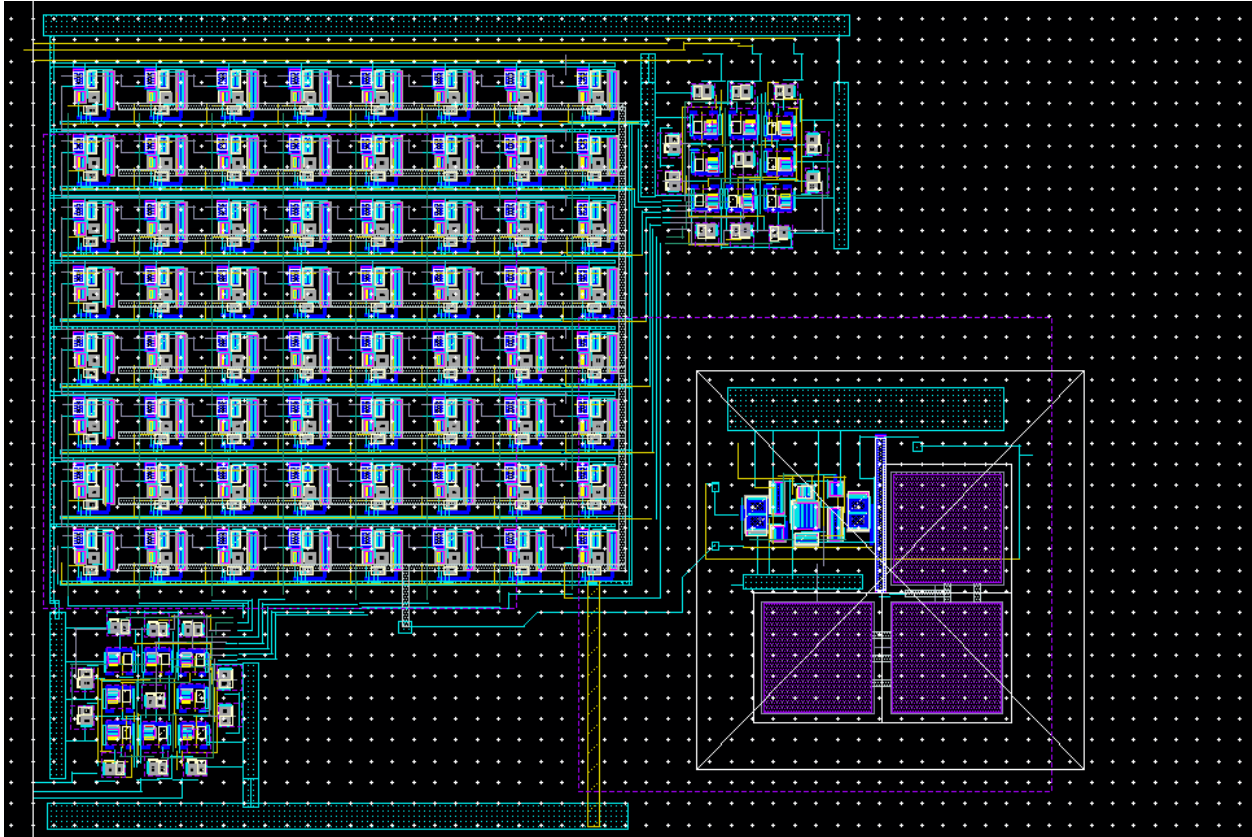
The ENOB distribution has a mean of 10.10 bits and a worst case of 8.69 bits, exceeding the 8-bit target by 0.69 bits even at the worst Monte Carlo iteration. The ENOB is computed from INL as $ENOB = 6 - \log_2(2 * INL_max_physical_LSB)$, which is equivalent to the standard definition. Values exceeding 8 bits indicate that the DAC's linearity corresponds to a precision finer than 8-bit resolution.

9.6 Yield Summary

Metric	Spec	Mean	Worst Case	Yield
INL_max	< 0.125 LSB	30.7 mLSB	~65 mLSB	100%
DNL_max	< 0.25 LSB	15.0 mLSB	~60 mLSB	100%
ENOB	≥ 8 bits	10.10 bits	8.69 bits	100%

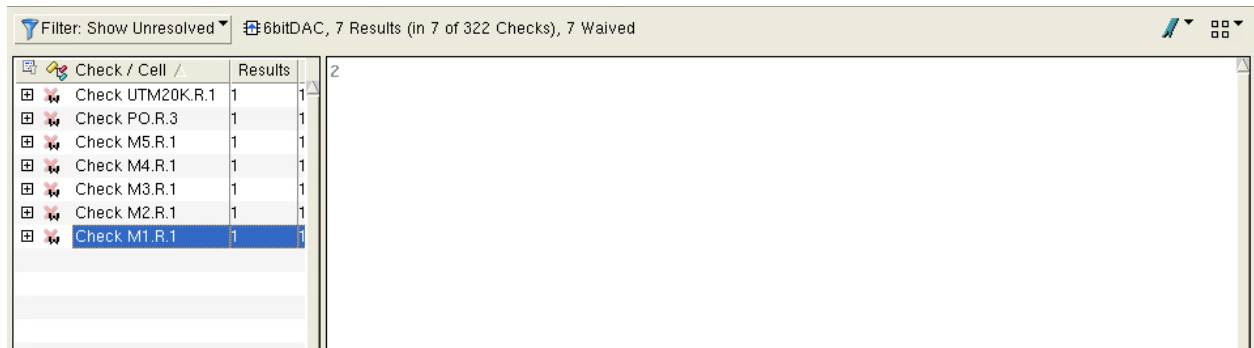
10. Final Layout

10.1 Layout



The completed final layout, which passes DRC and LVS. It is not super clean and not super well optimized, but it was still a lot of work and took a good amount of effort to get done even in its unclean state.

10.2 DRC



Check / Cell	Results
Check UTM20K.R.1	1
Check PO.R.3	1
Check M5.R.1	1
Check M4.R.1	1
Check M3.R.1	1
Check M2.R.1	1
Check M1.R.1	1

Clean DRC for final layout except for metal fill rules. My understanding is these shouldn't be too difficult to solve, and there is usually an automated tool to do so, but I'm unsure how.

10.3 LVS

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5 #####
6 ##                                ##
7 ##          C A L I B R E   S Y S T E M          ##
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9 ##          L V S   R E P O R T                   ##
10 ##                                ##
11 #####
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14 REPORT FILE NAME:          6bitDAC.lvs.report
15 LAYOUT NAME:              6bitDAC.sp ('6bitDAC')
16 SOURCE NAME:              6bitDAC.src.net ('6bitDAC')
17 RULE FILE:                _calibre.lvs
18 CREATION TIME:            Wed May 13 00:40:13 2026
19 CURRENT DIRECTORY:        /home/lhemming/TSMC180/LVS
20 USER NAME:                lhemming
21 CALIBRE VERSION:          v2025.2_14.11   Tue Apr 8 13:35:47 PDT 2025
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```

There are a couple of caveats to this completed LVS, one I still had some warnings describing invalid PATCHK requests, I was completely unsure what to do about these. The second major caveat is that I had to use a psub2 layer in order to not get a stamping error on softchk, since my opamp transistor bulk is connected to the substrate, and so is every other pmos on the chip there is a collision between the analog and digital vss bulk. My solution is not a real physical solution and rather just tells LVS not worry about it. My understanding is that solving this issue requires some other significant changes which I am unsure how to implement.

11. Challenges and Lessons Learned

11.1 VREF and Op-Amp Output Saturation

The original design used $V_{REF} = 3.3V$, which caused the op-amp output to saturate at the highest DAC codes. The issue manifested in the INL curve at codes near the top of the range. By plotting the mean INL across all Monte Carlo iterations, it was confirmed that this was a deterministic effect, the op-amp simply could not swing its output close enough to the positive supply rail. The fix was straightforward: lowering V_{REF} gave the op-amp sufficient output headroom. The final value of $1.0V$ was chosen to provide generous margin across process corners.

11.2 PDK Model Library Conflicts

Including both `cor_3v.scs` (section `tt_3v`) and `rf018.scs` (section `mc_3v`) simultaneously caused Cadence to report a duplicate subcircuit definition error for `nch3_mis` and `pch3_mis`. Both files define these mismatch-enabled 3V transistor models, and Spectre's parser does not allow redefinition. The solution was to disable `cor_3v.scs` when running Monte Carlo analysis, since `mc_3v` provides the mismatch variation needed for yield analysis.

11.3 Widlar Bias Stability Across Process Corners

The initial Widlar current reference caused approximately 15% of Monte Carlo iterations to fail catastrophically. The INL histogram was bimodal, with most iterations clustered near 30 mLSB and a secondary cluster at 5 to 20 LSB where the op-amp had completely lost its operating point. The root cause was that the degeneration resistor R_{REF} was too small, making the reference current overly sensitive to V_{TH} variation. At slow process corners, V_{TH} increased enough that the Widlar's current dropped below the minimum needed to sustain the op-amp's bias point. Increasing R_{REF} and widening the second-stage input transistor resolved the issue, producing a clean distribution and 100% yield.

11.4 Op-Amp Feedback Polarity

A persistent debugging issue was traced to the op-amp being wired in the wrong feedback configuration. The `vin+` and `vin-` pins were swapped, placing the op-amp in an inverting configuration rather than as a unity-gain buffer. This caused the DAC output to be non-

functional. I think it started when converting from the ideal design using the analog lib components, and then I just mis-wired and assumed it was correct for a long time. Other failures were blocking me from noticing this for a long time and a lot of time was lost here.

11.5 Row/Column Bit Assignment

The unit cell bus mapping assigns cell index k to row $(k \bmod 8)$ and column $(k \div 8)$. This means the row index changes with every increment of k (LSB behavior) while the column index changes every 8 increments (MSB behavior). Consequently, the row decoder must receive the least-significant code bits and the column decoder the most-significant bits. The initial assignment was reversed, producing a characteristic sawtooth transfer curve: V_{OUT} ramped correctly within each group of 8 codes (one column) but reset at each column boundary. A sawtooth pattern immediately identified the mapping error, and swapping the bit assignments produced the correct monotonic staircase.

12. Conclusion

A 6-bit R-string DAC achieving a mean ENOB of 10.1 bits (8-bit target) was designed and verified in the TSMC 180nm process. The design achieves 100% yield over 200 Monte Carlo iterations with both mismatch and process variation enabled, exceeding the 90% yield requirement. All analog components are implemented at the transistor level with no ideal elements except voltage sources. The final layout was able to pass DRC and LVS(except for metal density)

The design's robustness stems from three factors: deliberate over-sizing of the unit resistor, the inherent monotonicity of the R-string architecture (guaranteeing $DNL > -1$ LSB by construction), and the architecture's natural insensitivity to switch and op-amp mismatch (which contribute negligibly to static INL). These properties make the R-string an ideal choice for applications where yield under manufacturing variation is the primary design constraint.

Overall it was a very fun and interesting design to implement, honestly the layout wasn't that bad because it was very easy to copy and reuse some standard structures. If we had more stringent layout or area requirements there is a strong possibility I would have gone with a different design.